Harsh Karekar

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Professional Summary

Final-year ECE undergraduate with a strong foundation in AI, Machine Learning, and Data Science. Skilled in designing intelligent systems, analyzing complex datasets, and building scalable, production-ready solutions. Experienced in translating real-world problems into deployed applications. Seeking opportunities as an AI Engineer or ML Engineer.

Education

Vellore Institute of Technology | *Bhopal* — *2022 – 2026 (Expected)*
B.Tech in Electronics and Communication Engineering | CGPA: 8.39 / 10

Technical Skills

Languages: Python, Java, SQL
ML & Deep Learning: XGBoost, Random Forest, Autoencoder (TensorFlow/Keras), Scikit-learn, NLP, RAG Workflows, Agentic AI, LLMs
Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Jupyter
MLOps & Deployment: FastAPI, REST API Design, Model Serving, Cloud Deployment (Render, Vercel), Git/GitHub
Web & Tools: React.js, Node.js, Tailwind CSS

Projects

FraudShield | Live Demo | *Python, XGBoost, Autoencoders, TensorFlow/Keras, FastAPI, React* — *Aug – Nov 2025*

- Engineered a hybrid fraud detection system combining supervised XGBoost and an unsupervised Deep Autoencoder to detect known fraud signatures and zero-day anomalies across 284,807 transactions with only 0.17% fraud rate — a highly imbalanced real-world setting.
- Designed a Hybrid Risk Score ($\alpha\times$ XGBoost prob $+(1-\alpha)\times$ anomaly score, scaled 0–100%) with fraud flagging at >50% threshold, achieving strong generalization on unseen fraud patterns.
- Built and deployed a full-stack ML inference pipeline — FastAPI REST backend (Render) with batch inference and a React + TypeScript dashboard (Vercel) — serving real-time and batch fraud analysis.
- Handled extreme class imbalance by training the anomaly model exclusively on legitimate transactions, improving detection of evolving fraud not seen during supervised training.

PATLens – Placement Manager | *Python · LLM (Mistral) · Gmail API · Google Sheets API* — *Feb – May 2025*

- Built an end-to-end agentic AI pipeline that autonomously ingests unstructured placement emails from Gmail, extracts 15 structured fields (company, CTC, eligibility, deadlines, links) via LLM-based semantic parsing, and maintains a live Google Sheets tracker — zero manual effort.
- Replaced brittle regex extraction with LLM + deterministic post-processing, enabling reliable parsing of inconsistent human-written email formats with modular, cron-schedulable architecture.
- Designed an incremental append-only data pipeline with scheduled execution (cron/Task Scheduler), supporting scalable extension for eligibility checks and analytics.

Trader Performance vs Bitcoin Market Sentiment | *Python, Pandas, Scikit-learn, Streamlit* — *Dec 2025 – Feb 2026*

- Analyzed 10,000+ real trade executions from Hyperliquid DEX against the Bitcoin Fear & Greed Index, uncovering that traders are measurably more profitable during Fear regimes than Greed periods.
- Trained a Random Forest classifier on sentiment score, trade size, and direction (BUY/SELL) to predict trade-level profitability, demonstrating predictability using market sentiment as a feature.
- Built and deployed an interactive Streamlit dashboard with regime-based PnL charts, wallet leaderboard, BUY vs SELL profitability breakdown, and an AI-powered trade profitability predictor.

Certifications

- IBM Data Science Professional Certificate | *IBM / Coursera* — *July 2025*
- OCI Certified Generative AI Professional | *Oracle* — *Oct 2025*
- Generative AI using Watsonx | *IBM* — *April 2025*
- Artificial Intelligence Fundamentals | *IBM* — *Dec 2024*

Extracurricular

Technical Team Representative – Linux Club, VIT Bhopal — *Feb – Apr 2024*

- Represented the technical team in organizing a competitive **CTF event** focused on Linux fundamentals, terminal navigation, and cybersecurity challenges, managing event infrastructure for **150+ participants**.

Event Management Lead – Matrix, The Multimedia Club, VIT Bhopal — *Jan – Jun 2025*

- Led planning and execution of **2 major college-level multimedia events**, managing **500+ participants** and coordinating with **40+ cross-functional team members** to ensure smooth operations.